Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A body starts from rest and travels a distance x with uniform acceleration, then it travels a distance $2x$ with uniform speed, finally it travels a distance $3x$ with uniform retardation and comes to rest. If the complete motion of the particle is along a straight line, then the ratio of its average velocity to maximum velocity is

(A) $2/5$

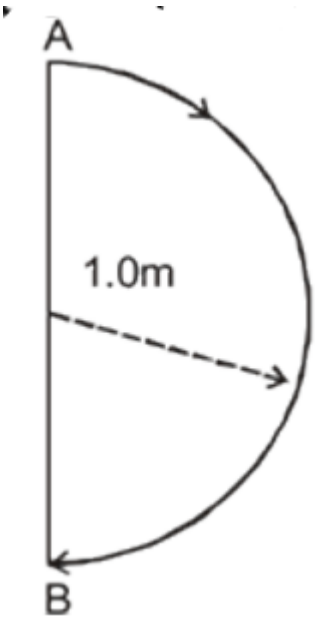
(B) $3/5$

(C) $4/5$

(D) $6/7$

Q.2

In 1.0 s, a particle goes from point A to point B, moving in a semicircle of radius 1.0 m (see Figure). The magnitude of the average velocity is



(A) 3.14 m/s

(B) 2.0 m/s

(C) 1.0 m/s

(D) Zero

Q.3

A man of height h walks in a straight path towards lamp post of height H with velocity v. Then velocity of the edge of the shadow on the ground will be

(A) $\frac{hv}{H+h}$

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(B) $\frac{\mathrm{H}v}{\mathrm{H}-h}$

(C) $\frac{\mathrm{H}+h}{\mathrm{H}v}$

(D) $\frac{(\mathrm{H}-h)}{\mathrm{H}h}$

Q.4

An athlete completes one round of a circular track of radius R in $40\mathrm{sec}$. What will be his displacement at the end of 3 min. $20\mathrm{sec}$?

(A) Zero

(B) $2\mathrm{R}$

(C) $2\pi R$

(D) $7\pi R$

Q.5

A body moves in straight line with velocity v_1 for $1/3^{\mathrm{rd}}$ time and for remaining time with v_2. Find average velocity.

(A) $\frac{\mathrm{v}_1}{3}+\frac{2\mathrm{v}_2}{3}$

(B) $\frac{\mathrm{v}_1}{3}+\frac{\mathrm{v}_2}{3}$

(C) $\frac{2v_1}{3}+\frac{v_2}{3}$

(D) $\mathrm{v}_1+\frac{2\mathrm{v}_2}{3}$

Q.6

A particle moves from $(2,3)\mathrm{m}$ to $(4,1)\mathrm{m}$. The magnitude of displacement is

(A) $2\ \mathrm{m}$

(B) $2\sqrt{3}\ \mathrm{m}$

(C) $2\sqrt{2}\ \mathrm{m}$

(D) $3\sqrt{2}\ \mathrm{m}$

Q.7

A man leaves his house for a cycle ride. He comes back to his house after half-an-hour after covering distance of one km. What is his average velocity for the ride?

(A) zero

(B) $2\ \mathrm{km\ h^{-1}}$

(C) $10\ \mathrm{km\ s^{-1}}$

(D) $\frac{1}{2}\ \mathrm{km\ s^{-1}}$

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Q.8

A point traversed half of the distance with velocity v0. The remaining part of the distance was covered with velocity v_1 & then velocity v_2 for equal time. The mean velocity of the point, averaged over the whole time of motion is

(A) $\frac{v_0+v_1+v_2}{3}$

(B) $\frac{2v_0+v_1+v_2}{3}$

(C) $\frac{v_0+.2v_1+2v_2}{3}$

(D) $\frac{2v_0(v_1+v_2)}{(2v_0+v_1+v_2)}$

Q.9

If a car covers $2/5^{\text{th}}$ of the total distance with v_1 speed and $3/5^{\text{th}}$ distance with v_2 then average speed is

(A) $\frac{1}{2}\sqrt{v_1 v_2}$

(B) $\frac{v_1+v_2}{2}$

(C) $\frac{2v_1 v_2}{v_1+v_2}$

(D) $\frac{5v_1 v_2}{3v_1+2v_2}$

Q.10

A person moves $30\ \text{m}$ north and then $20\ \text{m}$ towards east and finally $30\sqrt{2}\ \text{m}$ south-west direction. The displacement of the person from the origin will be

(A) $10\ \text{m}$ along north

(B) $10\ \text{m}$ along south

(C) $10\ \text{m}$ along west

(D) zero

Q.11

A bus travelling the first one third distance at a speed of $10\ \text{km/h}$, the next one third at $20\ \text{km/h}$ and the last one-third at $60\ \text{km/h}$. The average speed of the bus is

(A) $9\ \text{km/h}$

(B) $16\ \text{km/h}$

(C) $18\ \text{km/h}$

(D) $48\ \text{km/h}$

Q.12

A car moves with a speed of $60\ \text{km/hr}$ from point A to point B and then with the speed of $40\ \text{km/hr}$ from point B to point C. Further it moves to point D with a speed equal to its average speed between A and C. Points

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A, B, C and D are collinear and equidistant. The average speed of the car between A and D is

(A) $30\ \mathrm{km/hr}$

(B) $50\ \mathrm{km/hr}$

(C) $48\ \mathrm{km/hr}$

(D) $60\ \mathrm{km/hr}$

Q.13

Let us consider a boat which moves with a velocity $v_{bw} = 5\ \mathrm{km\ h^{-1}}$ relative to water. At time $t = 0$, the boat passes through a piece of cork floating in water while moving downstream. If it turns back at time $t = t_1$, when and where does the boat meet the cork again? Assume $t_1 = 30\ \mathrm{min}$.

(A) 45 mins

(B) 30 mins

(C) 2 hr

(D) 1 hr

Q.14

On a two lane road, car A is travelling with a speed of $36\mathrm{kmh}^{-1}$. Two cars B and C approach car A in opposite directions with a speed of $54\ \mathrm{km\ h^{-1}}$. At a certain instant, when the distance AB is equal to AC, both $1\ \mathrm{km}$, B decided to overtake A before C does. What minimum acceleration of car B is required to avoid an accident?

(A) $5\ \mathrm{ms^{-2}}$

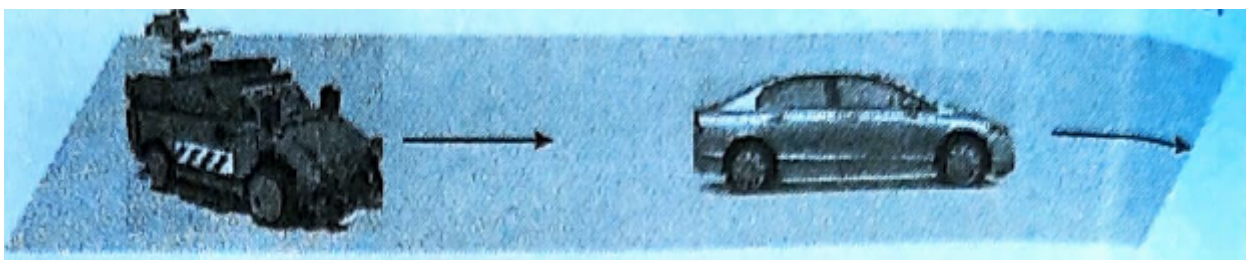
(B) $10\ \mathrm{ms^{-2}}$

(C) $6\ \mathrm{ms^{-2}}$

(D) $1\ \mathrm{ms^{-2}}$

Q.15

A police van moving on a highway with a speed of $30\ \mathrm{km}_h^{-1}$ fires a bullet at a thief's car speeding away in a same direction with a speed of $192\ \mathrm{km\ h^{-1}}$. If the muzzle speed of the bullet $150\ \mathrm{m\ s^{-1}}$, with what speed does the bullet hit the thief's car?



(A) $105\ \mathrm{ms^{-1}}$

(B) $110\ \mathrm{ms^{-1}}$

(C) $106\ \mathrm{ms^{-1}}$

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(D) $111\ \mathrm{ms}^{-1}$

Answers & Solutions

Q.1 Answer:

$3/5$

Solution:

Case I: Body starts from rest, $u = 0$.

uniform acceleration, $a = \alpha$.

travels distance $= x$.

Let t_1 **= time taken to travel** x **distance;**

Now, using equation of motion;

$$v = u + at$$

$$v' = 0 + \alpha t_1$$

$$\alpha = \frac{v'}{t_1}$$

Also,

$$s = ut + \tfrac{1}{2}at^2.$$

$$x = 0 + \tfrac{1}{2}\alpha t_1^2.$$

$$x = \tfrac{v'}{t_1}\left(t_1^2\right) = \tfrac{v't_1}{2} \Rightarrow t_1 = \tfrac{2x}{v'}$$

Now, Case II body travels $2x$ **distance with uniform velocity so acceleration = 0, So in time** t_2 **we have,**

$$s = ut + \frac{1}{2}at^2$$

$$2x = v't_2^n$$

$$t_2 = \frac{2x}{v'}$$

Now, Case III body starts retard. So,

$$\text{retardation } = \beta, \text{ distance } = 3x, \text{ time } = t_3.$$

$$v = u + at$$

$$0 = v' + (-\beta)t_3.$$

$$\beta = \tfrac{v'}{t_3}.$$

Also,

$$s = ut + \frac{1}{2}at^2$$

$$3x = v't_3 - \frac{\beta}{2}t_3^2$$

$$\text{or,} \quad t_3 = \frac{6x}{v'} \quad \left(\beta = \frac{v'}{t_3}\right).$$

$$\text{Now, average velocity } = \tfrac{\text{total distance}}{\text{total time}} = \tfrac{x+2x+3x}{t_1+t_2+t_3}$$

$$v_{av} = \tfrac{6x}{\frac{2x}{v'}+\frac{2x}{v'}+\frac{6x}{v'}} = \tfrac{3v'}{5}$$

Clearly, moximum velocity will be V'**. so,** $V_{\max} = V$ **'.**

Thus

$$\text{ratio } = \tfrac{v_{av}}{v'} = \tfrac{3v'}{5} \times \tfrac{1}{v'} = \tfrac{3}{5}$$

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Hence, we get ratio $= 3 : 5$

Q.2 Answer:

$2.0\ \mathrm{m/s}$

Solution:

$$|\ \text{Average velocity}\ | = \frac{|\ \text{displacement}\ |}{\text{time}}$$

$$= \frac{2r}{t} = 2 \times \frac{1}{1} = 2\ \mathrm{m/s}$$

Q.3 Answer:

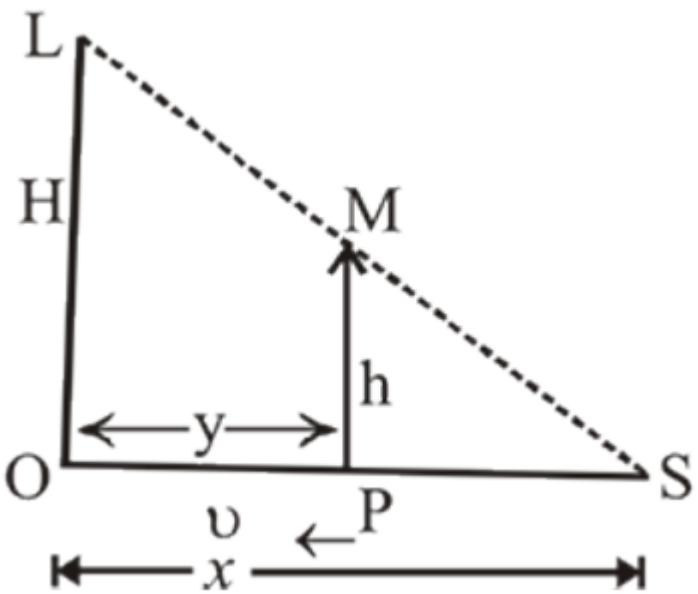
$\frac{\mathrm{H}v}{\mathrm{H}-h}$

Solution:

as ΔLOS **and** ΔMPS **are similar so,** $\frac{\mathrm{H}}{x} = \frac{\mathrm{h}}{x-y} \Rightarrow (\mathrm{H}-\mathrm{h})x = \mathrm{H}y$

$\Rightarrow (\mathrm{H}-\mathrm{h})\frac{\mathrm{dx}}{\mathrm{dt}} = \mathrm{H}\frac{\mathrm{dy}}{\mathrm{dt}} = \mathrm{Hv}$

$\Rightarrow \mathrm{V_s} = \frac{\mathrm{Hv}}{\mathrm{H}-\mathrm{h}}$



Q.4 Answer:

Zero

Solution:

Total time of motion is $3\ \mathrm{min}\ 20\mathrm{sec} = 200\mathrm{sec}$**. As time period of circular motion is** $40\mathrm{sec}$ **so in** $200\mathrm{sec}$ **athlete will complete revolution i.e., he will be at starting point**

i.e., displacemet = zero.

Q.5 Answer:

$\frac{\mathrm{v}_1}{3} + \frac{2\mathrm{v}_2}{3}$

Solution:

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$$v_{av} = \frac{v_1 t/3 + v_2(2t/3)}{t} = \frac{v_1}{3} + \frac{2v_2}{3}$$

Q.6 Answer:

$2\sqrt{2}\ \text{m}$

Solution:

Displacement $= \sqrt{(4-2)^2 + (3-1)^2} = \sqrt{4+4} = 2\sqrt{2}\ \text{m}$

Q.7 Answer:

zero

Solution:

Since displacement is zero.

Q.8 Answer:

$\frac{2v_0(v_1+v_2)}{(2v_0+v_1+v_2)}$

Solution:

Let the total distance be d**. Then for first half distance, time** $= \frac{d}{2v_0}$**, next distance.** $= v_1 t$ **and last half distance** $= v_2 t$

$\therefore \quad v_1 t + v_2 t = \frac{d}{2}; t = \frac{d}{2(v_1+v_2)}$

Now average speed $t = \frac{d}{\frac{d}{2v_0} + \frac{d}{2(v_1+v_2)} + \frac{d}{2(v_1+v_2)}}$

$= \frac{2v_0(v_1+v_2)}{(v_1+v_2)+2v_0}$

Q.9 Answer:

$\frac{5v_1v_2}{3v_1+2v_2}$

Solution:

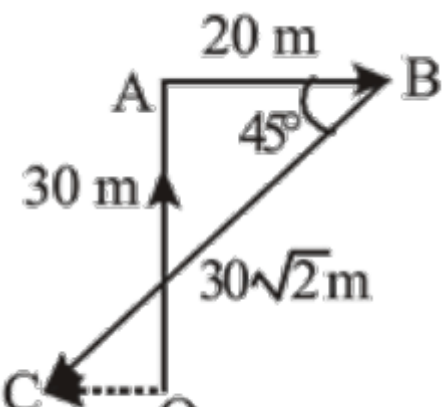
Average speed $= \frac{\text{Total distance travelled}}{\text{Total time taken}}$

$= \frac{x}{\frac{2x/5}{v_1} + \frac{3x/5}{v_2}} = \frac{5v_1v_2}{3v_1+2v_2}$

Q.10 Answer:

$10\ \text{m}$ along west

Solution:



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From figure, $\overrightarrow{OA} = 0\hat{i} + 30\hat{j}$

$\overrightarrow{AB} = 20\hat{i} + 0\vec{j}$

$\overrightarrow{BC} = -30\sqrt{2}\cos 45^\circ \hat{i} - 30\sqrt{2}\sin 45^\circ \hat{j}$

$= -30\hat{i} - 30\vec{j}$

$\therefore$ **Net displacement,** $\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AB} + \overrightarrow{BC} = -10\hat{i} + 0\hat{j}$

$|\overrightarrow{OC}| = 10$ m

Q.11 Answer:

18 km/h

Solution:

Average speed $= \frac{s}{\frac{s/3}{10} + \frac{s/3}{20} + \frac{s/3}{60}} = \frac{s}{s/18} = 18$ km/h

Q.12 Answer:

48 km/hr

Solution:

Let the points A, B, C **and** D **be separated by** 1 km. **Then** $t_{AB} = \frac{1}{60}hr, t_{BC} = \frac{1}{40}hr$

$\therefore < v_{AC} > = \frac{1+1}{\frac{1}{60}+\frac{1}{40}} = 48$ km/hr $\Rightarrow t_{CD}\frac{1}{48}hr$

Now $< v_{AD} > = \frac{1+1+1}{\frac{1}{60}+\frac{1}{40}+\frac{1}{48}} = 48$ km/hr

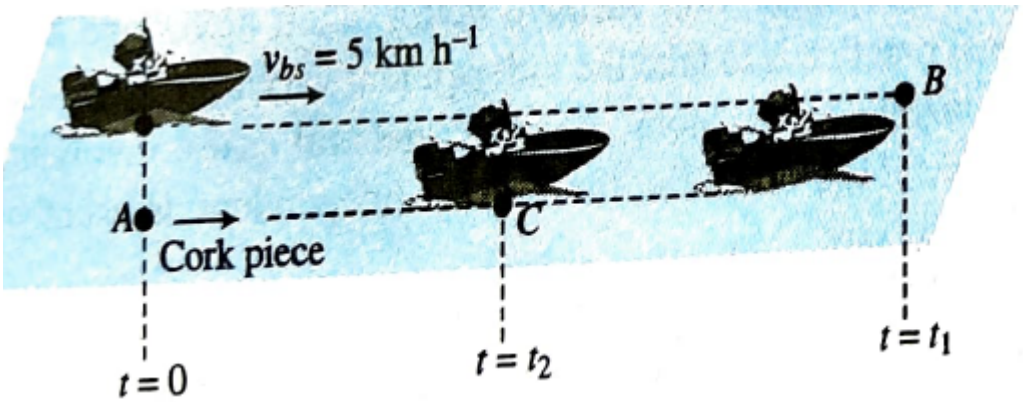
Q.13 Answer:

1 hr

Solution:

Time of travelling of boat from A **to** B (t_1) **and then** B **to** C

(t_1') = **time of moving the cork from** A **to** C.



Velocity of boat from A **to** B

$\vec{v}_{b,w} + \vec{v}_w = (5 + u)\text{kmh}^{-1}$

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And velocity of boat from B to C

$\vec{v}_{b,w} + \vec{v}_w = (5-u)\text{kmh}^{-1}$

Distance moved by boat in time t_1

$AB = (5+u)t_1$

And distance moved by boat in time $t_1' = \text{BC} = (5-u)t_1'$

Distance moved by cork during this time

$AC = u\,(t_1 + t_1')$

But $AB = AC + BC$

$(5+u)t_1 = u\,(t_1 + t_1') + (5-u)t_1'$

$5t_1 = 5t_1' \Rightarrow t_1' = t_1 = 30 \text{ min}$

Hence, the cork meets the boat again after 1 h

Q.14 Answer:

1 ms^{-2}

Solution:

At the instant when car B decides to overtake car A, the velocities of cars are:

$v_A = 36 \times \frac{5}{18} = 10 \text{ m s}^{-1}$

$v_B = 54 \times \frac{5}{18} = 15 \text{ m s}^{-1}$

and $v_C = -54 \times \frac{5}{18} = -15 \text{ m s}^{-1}$

Velocity of car B relative to A,

$\vec{v}_{BA} = \vec{v}_B - \vec{v}_A = 15 - 10 = 5 \text{ m s}^{-1}$

Velocity of car C relative to A,

$\vec{v}_{CA} = \vec{v}_C - \vec{v}_A = -15 - 10 = -25 \text{ m s}^{-1}$

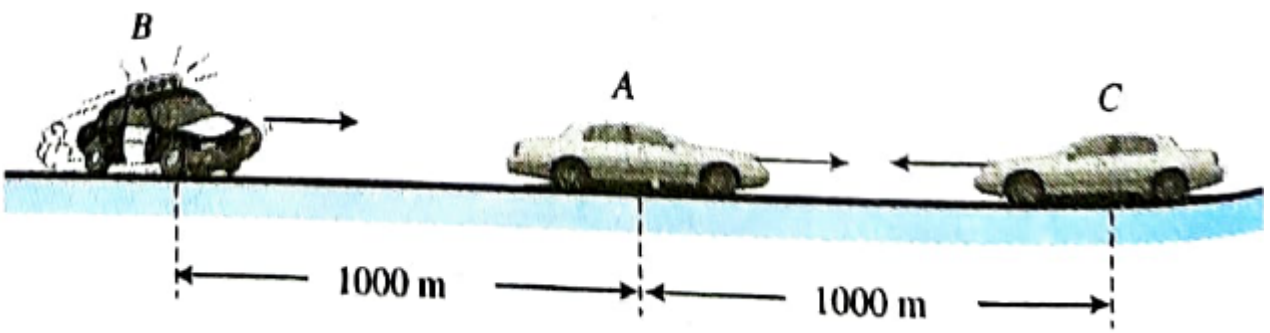
Time required by car C to just cross A $= \frac{1000}{v_{CA}} = \frac{1000}{25} = 40 \text{ s}$

In order to avoid accident, car B must overtake A in this time. So

$1000 = v_{BA}t + \frac{1}{2}a_{BA}t^2$

$1000 = 5 \times 40 + \frac{1}{2}a_{BA} \times 40^2$

$a_{81} = 1 \text{ m s}^{-2}$**. acceleration that car B requires to avoid accident.**



Q.15 Answer:

105 ms^{-1}

Solution:

Speed of police van $= 30 \times \frac{5}{18} = \frac{25}{3} \text{ m s}^{-1}$

The muzzle velocity, that is, the velocity of bullet with respect to van is

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$$[\vec{v}_{\text{bullet}}]_{\text{van}} = [\vec{v}_{\text{bullet}}]_{\text{ground}} - [\vec{v}_{\text{van}}]_{\text{ground}}$$

$$[\vec{v}_{\text{bullet}}]_{\text{ground}} = [\vec{v}_{\text{bullet}}]_{\text{van}} + [\vec{v}_{\text{van}}]_{\text{ground}}$$

$$= 150 + \frac{25}{3} = \frac{475}{3}\ \text{m s}^{-1}$$

Speed of thief's car $= 192 \times \frac{5}{18} = \frac{160}{3}\ \text{m s}^{-1}$

Now velocity of bullet with respect to the thief's car

$$[\vec{v}_{\text{bullet}}]_{\text{car}} = [\vec{v}_{\text{bullet}}]_{\text{ground}} - [\vec{v}_{\text{car}}]_{\text{ground}}$$

$$= \frac{475}{3} - \frac{160}{3} = 105\ \text{m s}^{-1}$$

Hence, the bullet hits the thief's car with speed $105\ \text{m s}^{-1}$**.**

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